

Spotify Music Analytics Dashboard

Power BI Project Documentation

Spotify Music Analytics Dashboard built in Power BI to analyze songs, artists, albums, popularity, duration, rankings, release years, and monthly trends with an interactive Spotify-style UI.

Displayed KPI	Value
Distinct Songs	789
Artists	342
Average Popularity	90
Average Duration	3.28 minutes

Spotify Music Analytics Dashboard — Project Overview

1. Project Summary

The Spotify Music Analytics Dashboard is a Power BI-based data analytics project designed to convert music data into an interactive, visually engaging dashboard.

The project combines analytical KPIs with a Spotify-inspired interface so that users can move from a high-level summary to artist- and song-level details.

2. Dataset Snapshot

The dashboard screenshots show a music dataset containing information related to:

- Songs
- Artists
- Albums
- Album type
- Popularity
- Release year
- Month
- Duration
- Records/chart-related metrics

- Explicit/non-explicit classification

The displayed dashboard state includes examples such as:

These values represent the state visible in the supplied dashboard screenshots and may change when filters or the underlying dataset change.

3. Dashboard Architecture

Home Page

A branded landing page that introduces the project and provides navigation to the analytical pages.

Overview Page

Designed for management-level analysis. It combines KPI cards, categorical breakdowns, time trends, and top-song analysis.

Artists Page

Designed to compare artists using song counts, album counts, compilation records, popularity, duration, and other artist-level metrics.

Songs Page

Designed for detailed song-level exploration, ranking, popularity, duration, and record-related metrics.

4. Main Visuals

KPI Cards

Used for quick identification of:

- Distinct songs
- Artists
- Average popularity
- Average duration

Bar Charts

Used for:

- Songs by artist
- Total popularity by songs
- Albums by artist
- Compilation records by song

- Artist/song comparisons

Donut Charts

Used for:

- Album type
- Explicit vs non-explicit songs
- Songs by year
- Average popularity by album type

Line Chart

Used to show Average Popularity by Month, helping identify monthly changes in music popularity.

Column Chart

Used to show Distinct Songs by Month, highlighting monthly volume patterns.

Detail Tables

Used to provide granular song/artist metrics, including:

- Maximum popularity
- Average popularity
- Average duration
- Total records
- Total albums
- Average position
- Highest popularity score
- Shortest song duration
- Worst position
- Single records

5. Analytical Story

The dashboard is structured as a progression:

Discover → Compare → Investigate

1. Discover: Start with the Home and Overview pages to understand the dataset.
2. Compare: Use artist and song charts to identify leading performers and tracks.
3. Investigate: Use detailed tables and ranking metrics to inspect individual songs and artists.

6. Key Insights Supported by the Dashboard

The dashboard makes it possible to identify:

- Artists with the largest number of songs.
- Songs with very high popularity.
- Differences between album and single records.
- Explicit vs non-explicit content distribution.
- Monthly changes in popularity.
- Monthly changes in the number of distinct songs.
- Artists with multiple albums.
- Songs with strong popularity scores.
- Songs with shorter or longer average durations.
- Ranking and record-related performance.

7. Power BI Skills Demonstrated

This project is suitable as a portfolio example for demonstrating:

- Power Query data preparation
- Data modeling
- DAX measures
- KPI development
- Aggregation and ranking
- Time intelligence concepts
- Interactive visualizations
- Report navigation
- Dashboard UX/UI design
- Data storytelling

8. Design Approach

The interface uses a dark background with Spotify-green highlights to create a recognizable music-analytics experience.

The design includes:

- Consistent navigation
- Rounded cards
- Album artwork

- Music-player visual
- Compact KPI panels
- High-contrast analytical charts
- Detailed tables
- Scrollable visual areas

9. Recommended GitHub README Sections

For the GitHub repository, the recommended order is:

1. Project title
2. One-line description
3. Dashboard preview
4. Project overview
5. Objectives
6. Dataset
7. Dashboard pages
8. KPIs and metrics
9. Tools and technologies
10. Business questions
11. Repository structure
12. How to use
13. Portfolio skills
14. Author

10. Short Portfolio Description

Spotify Music Analytics Dashboard | Power BI

An interactive Spotify-inspired Power BI dashboard for analyzing music data across songs, artists, albums, popularity, duration, rankings, content type, and time-based trends. The project combines DAX-driven KPIs, interactive charts, detailed tables, and a custom dark Spotify-style UI to deliver an engaging data storytelling experience.

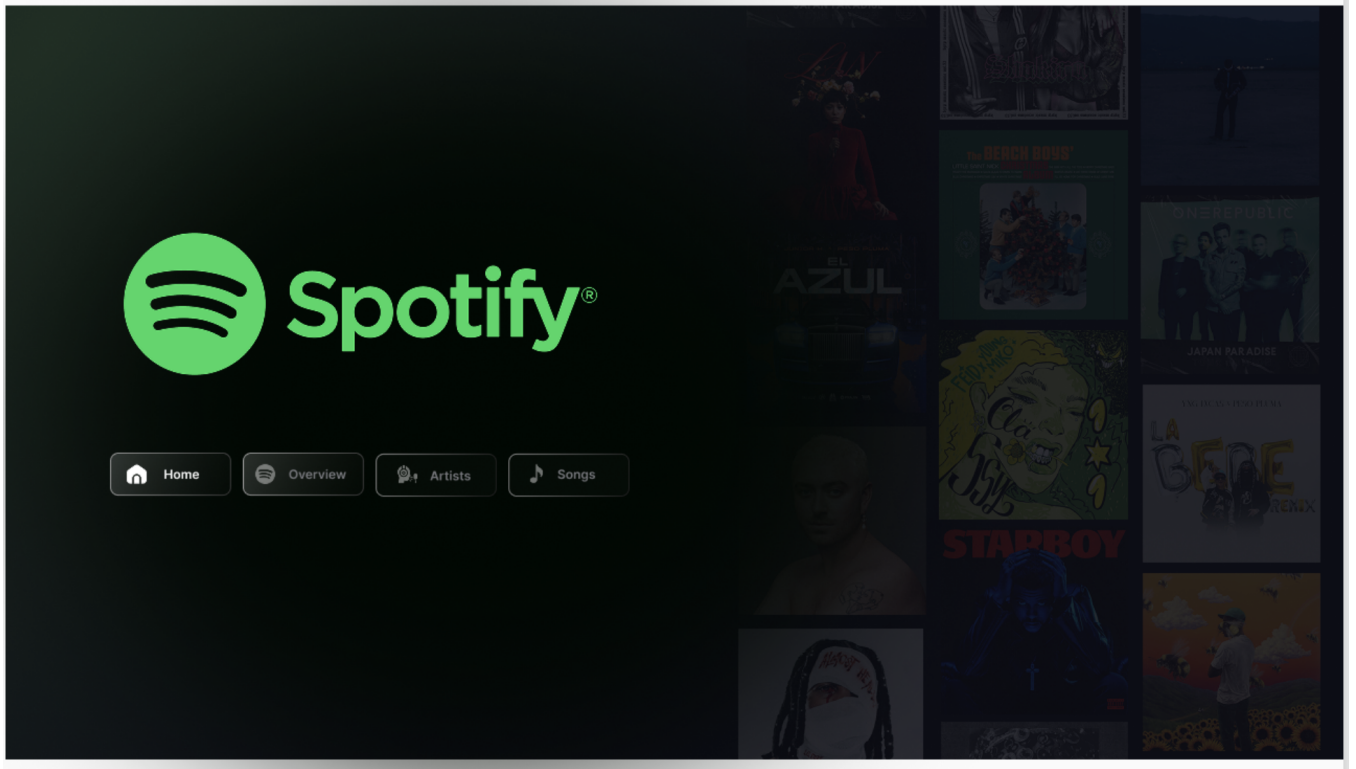
11. GitHub Repository Description

Spotify Music Analytics Dashboard built in Power BI to analyze songs, artists, albums, popularity, duration, rankings, release years, and monthly trends with an interactive Spotify-style UI.

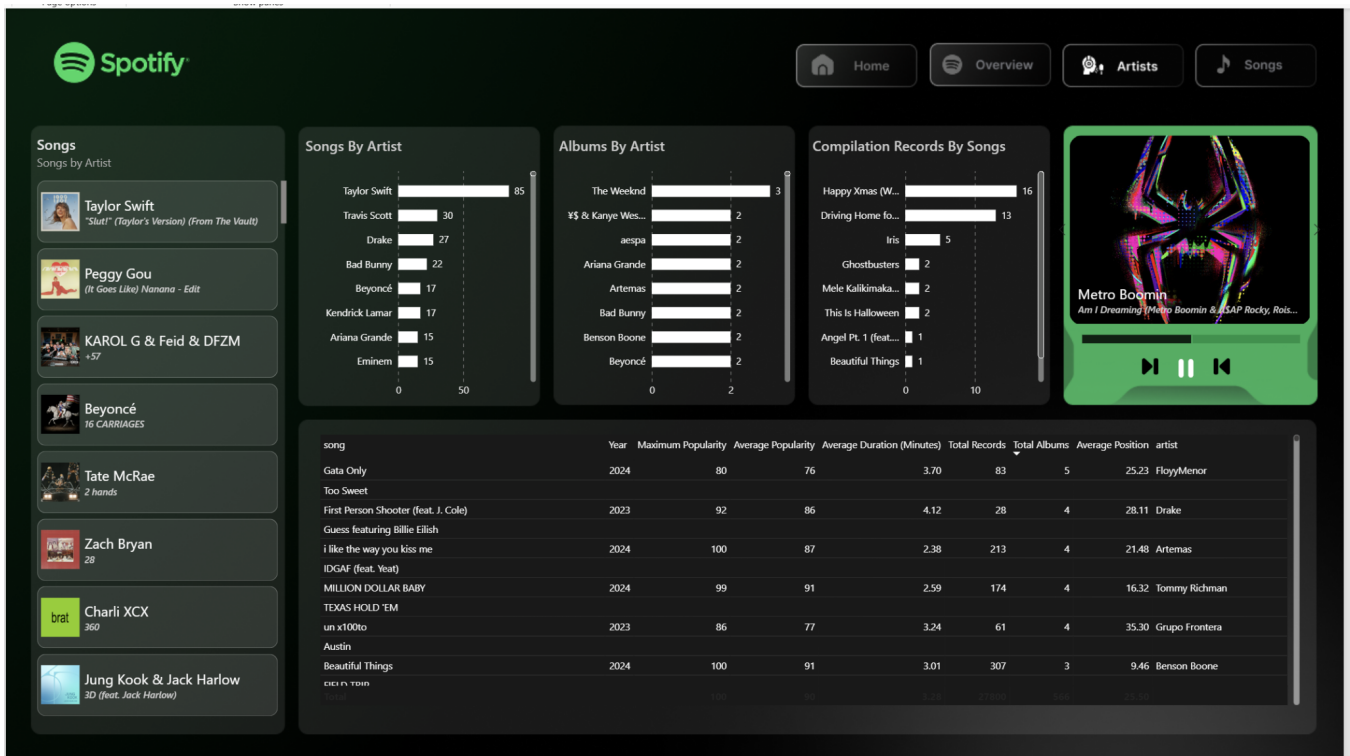
Dashboard Screenshots

The following pages contain the four supplied dashboard screenshots used as the visual reference for this documentation.

Dashboard Screenshot 1



Dashboard Screenshot 3



Dashboard Screenshot 4

